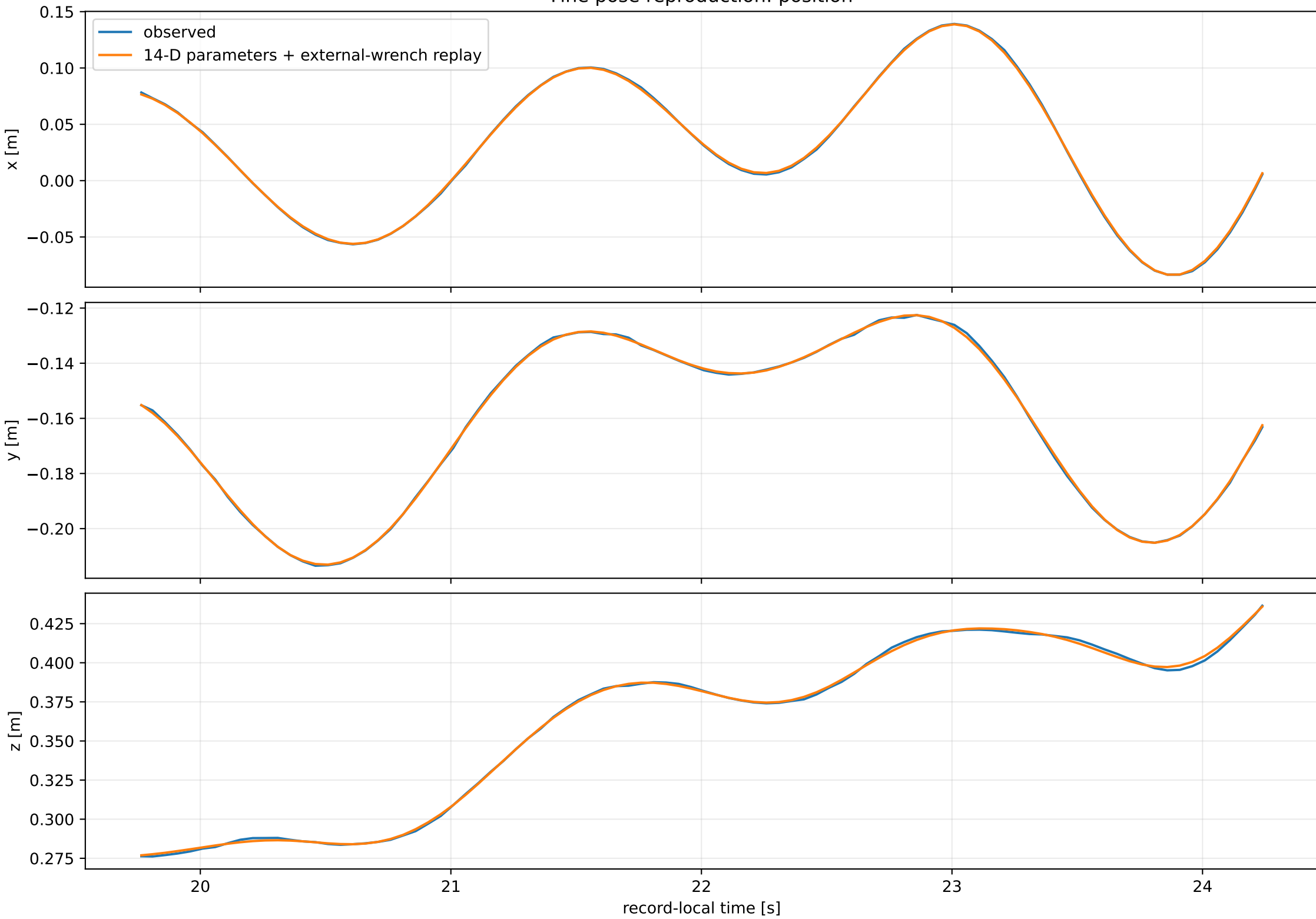
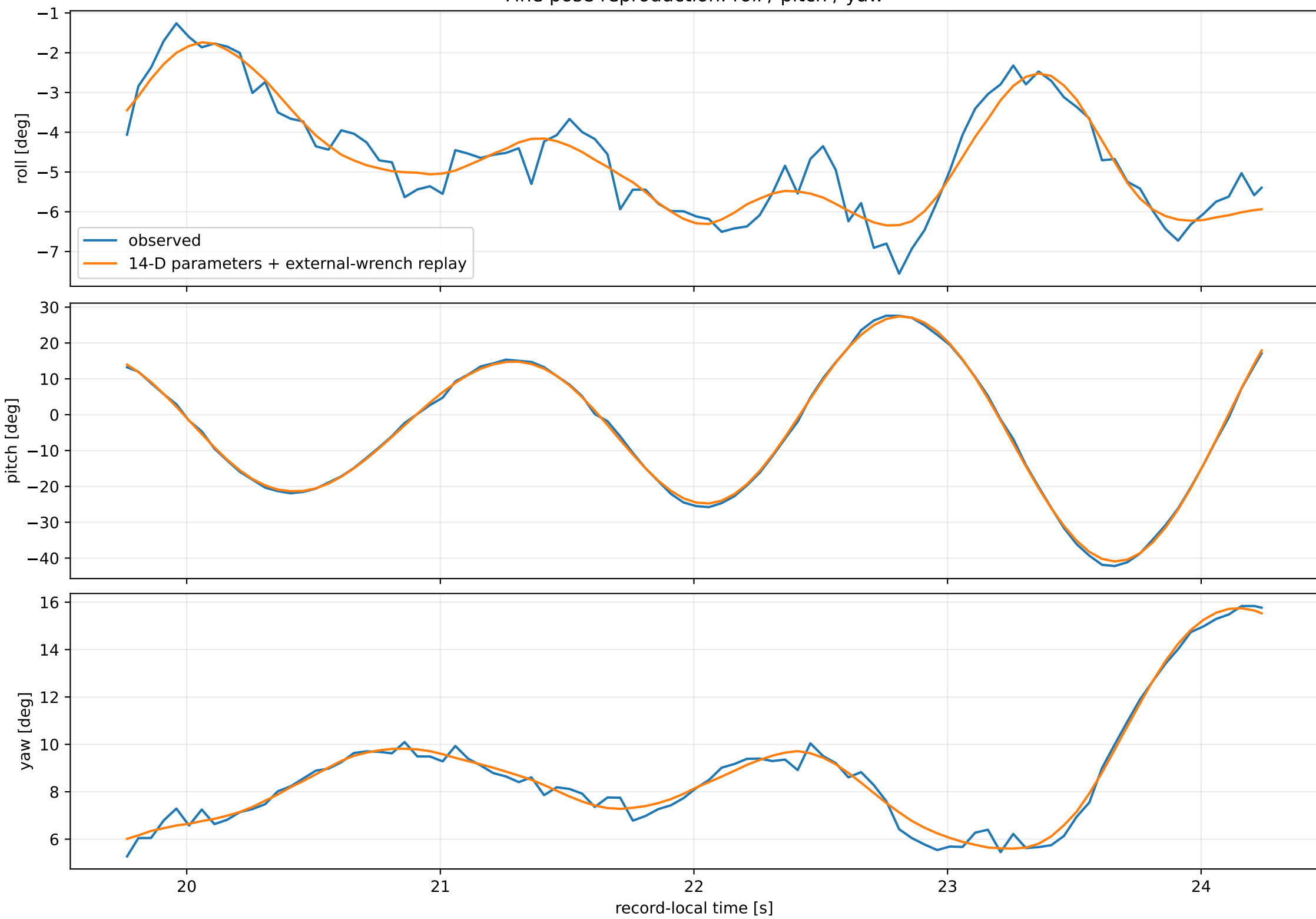


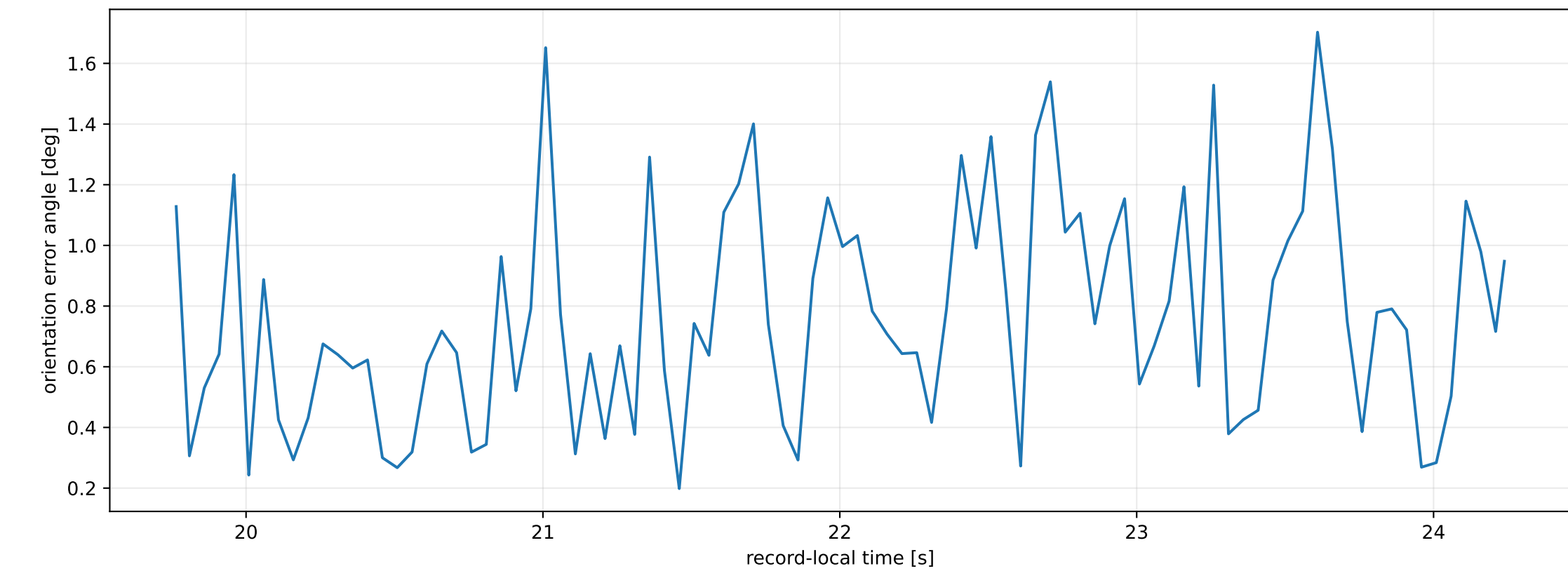
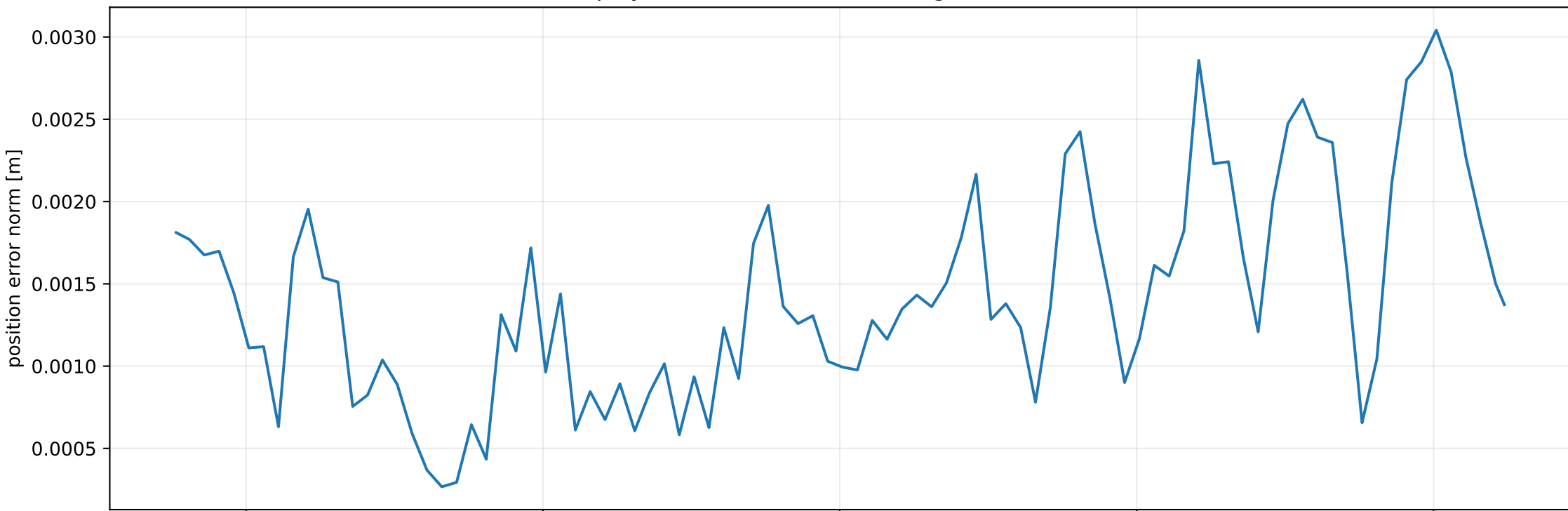
Fine pose reproduction: position



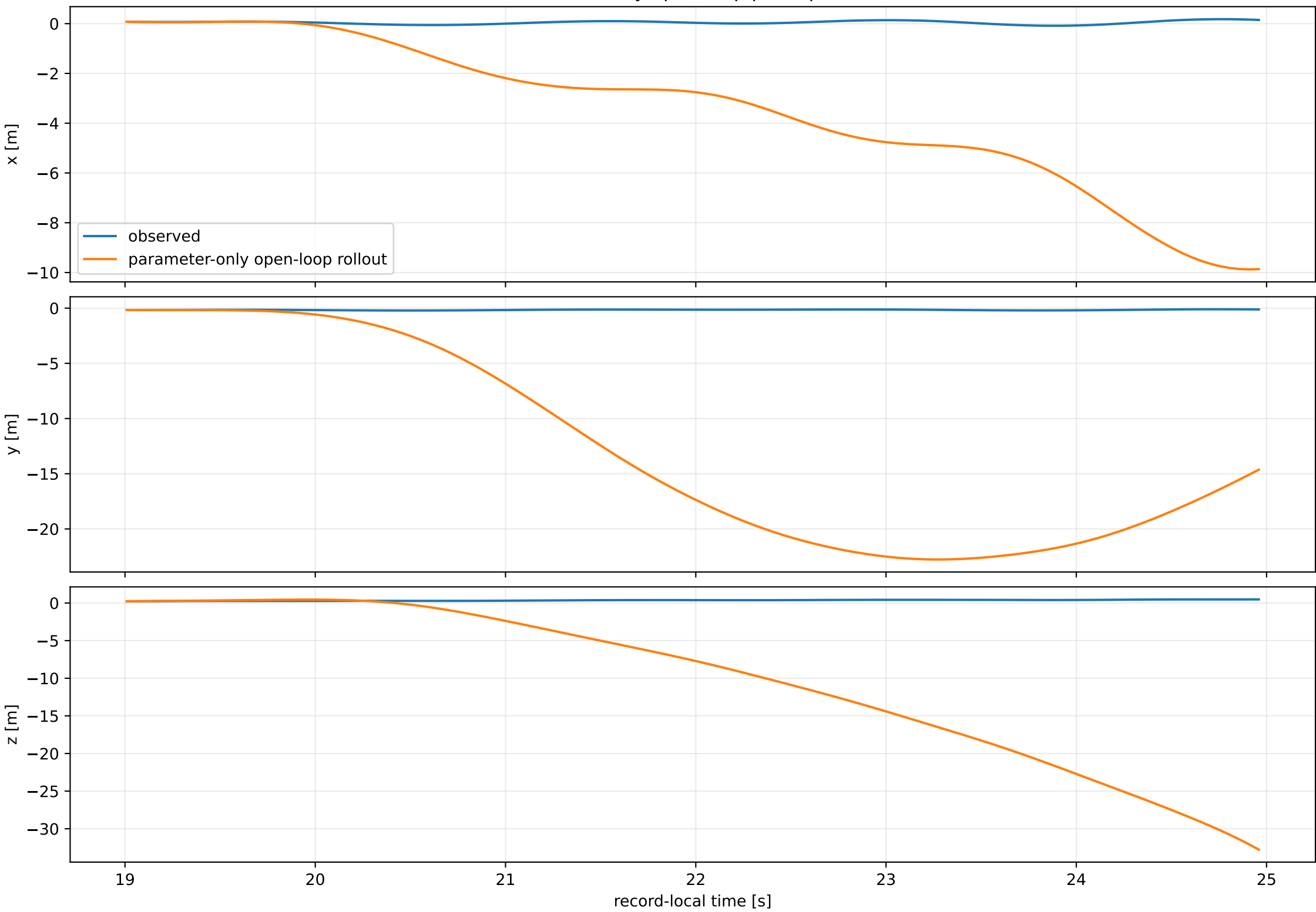
Fine pose reproduction: roll / pitch / yaw



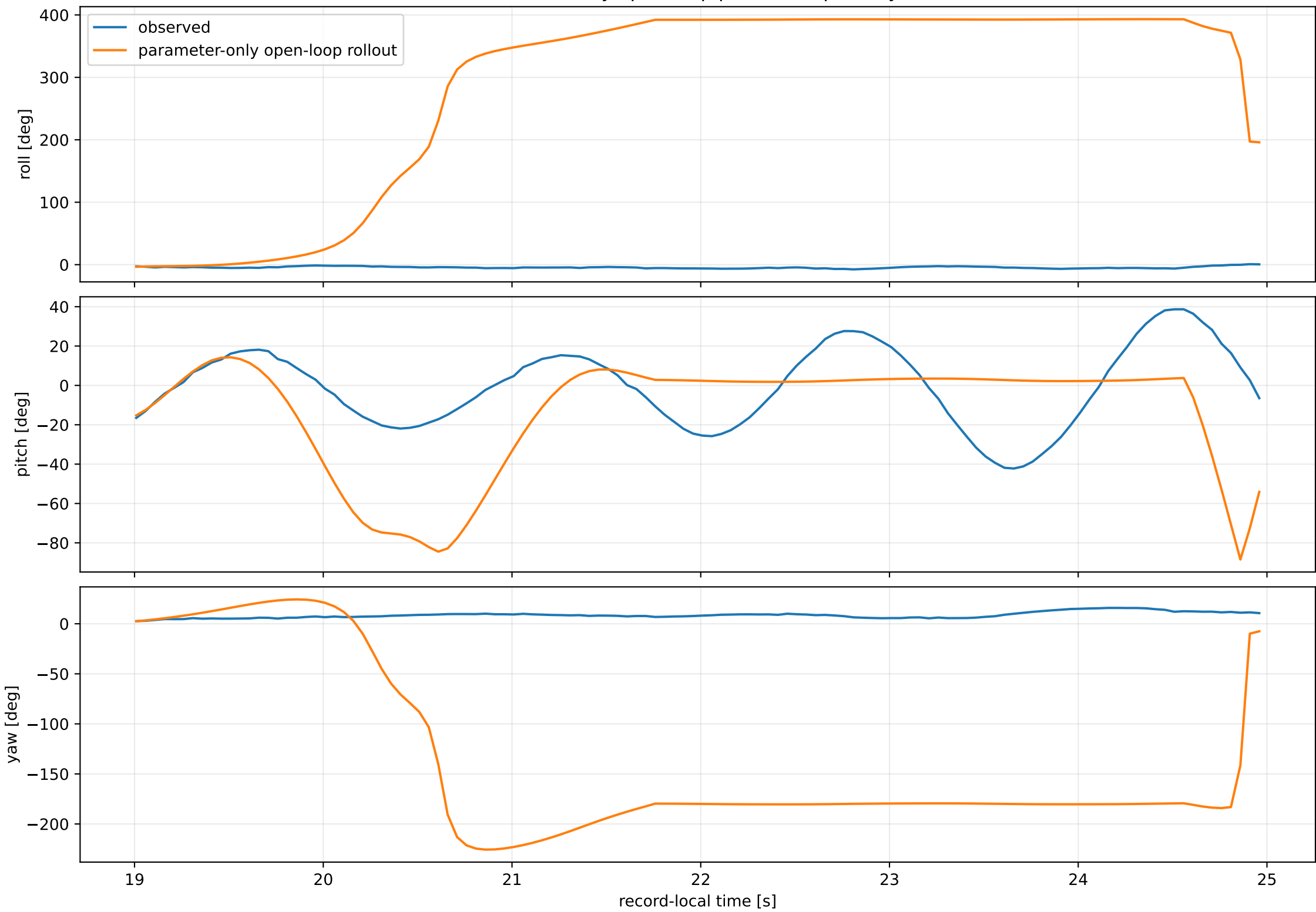
Fine replay error after the 14-D change — lower is better



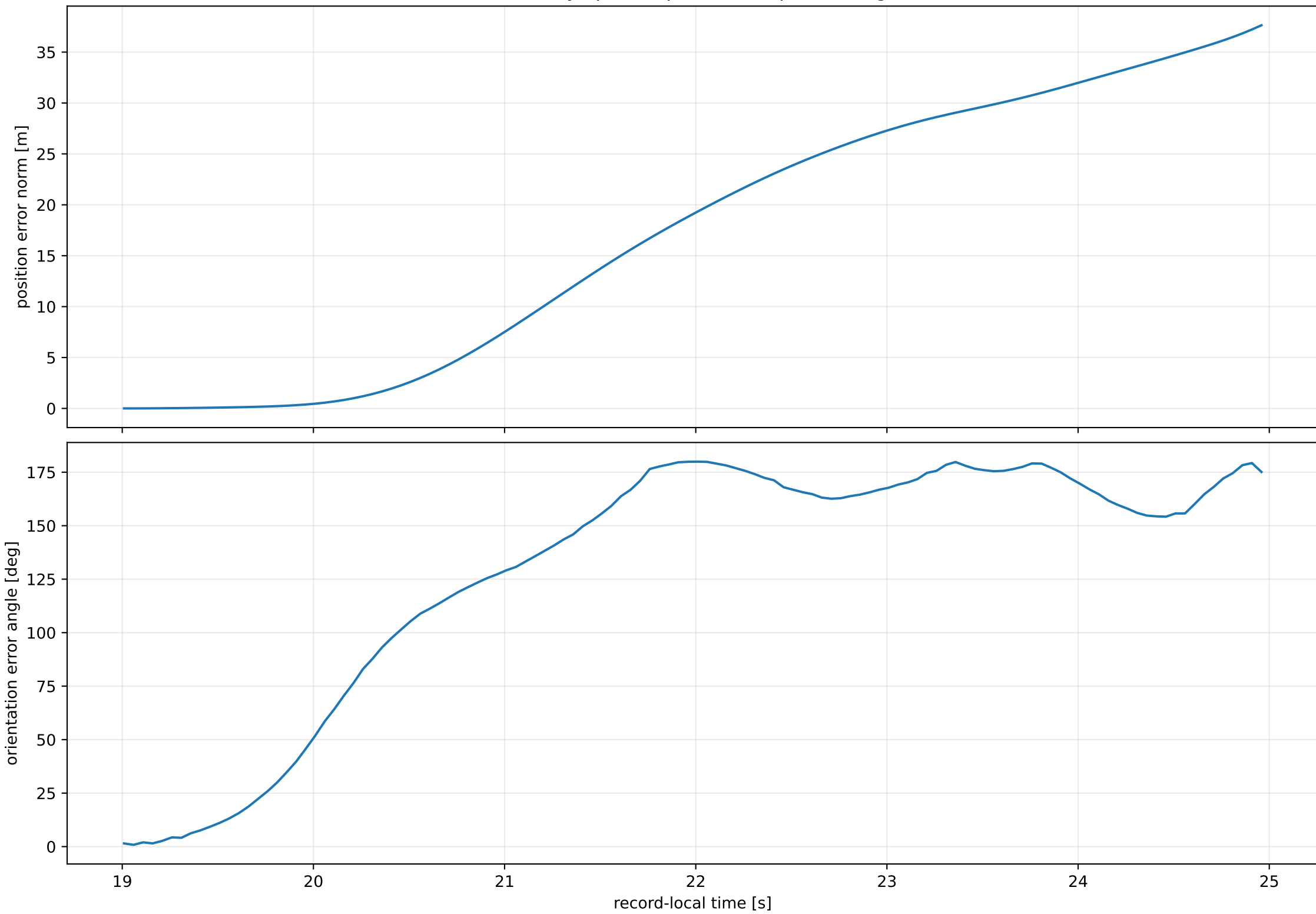
Parameter-only open-loop pose: position



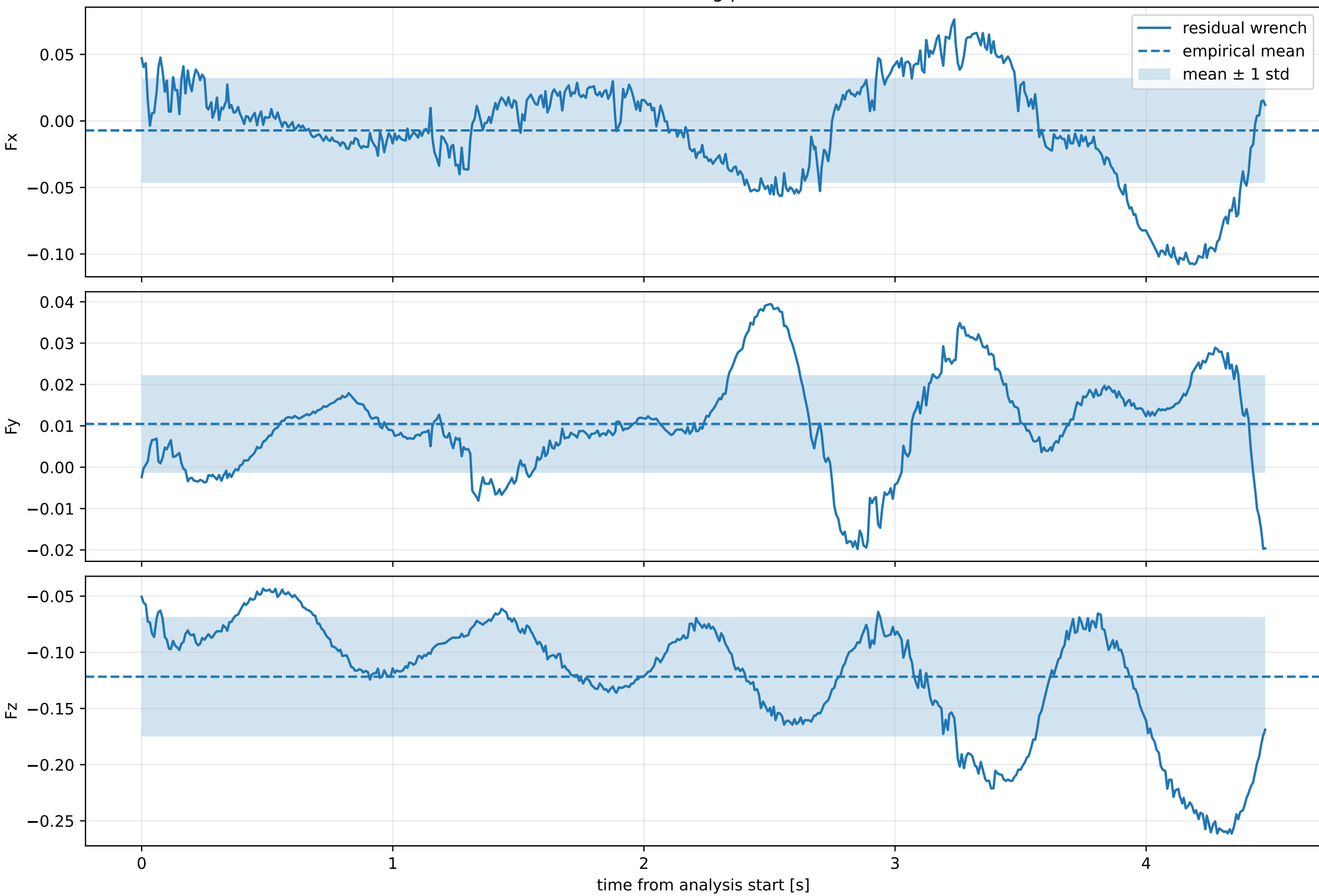
Parameter-only open-loop pose: roll / pitch / yaw



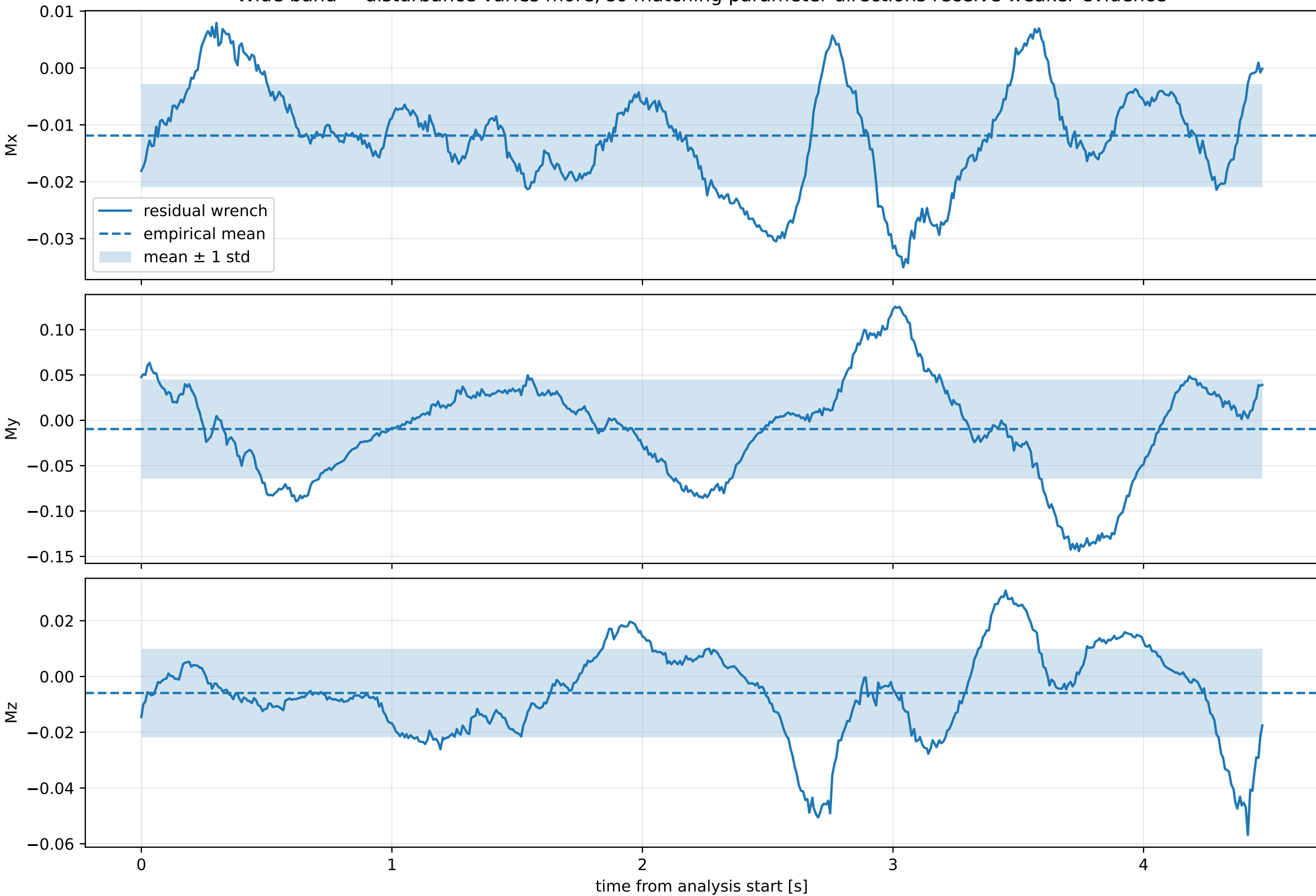
Parameter-only open-loop rollout (separate diagnostic)



Dimensionless residual force: empirical mean  $\pm$  1 std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

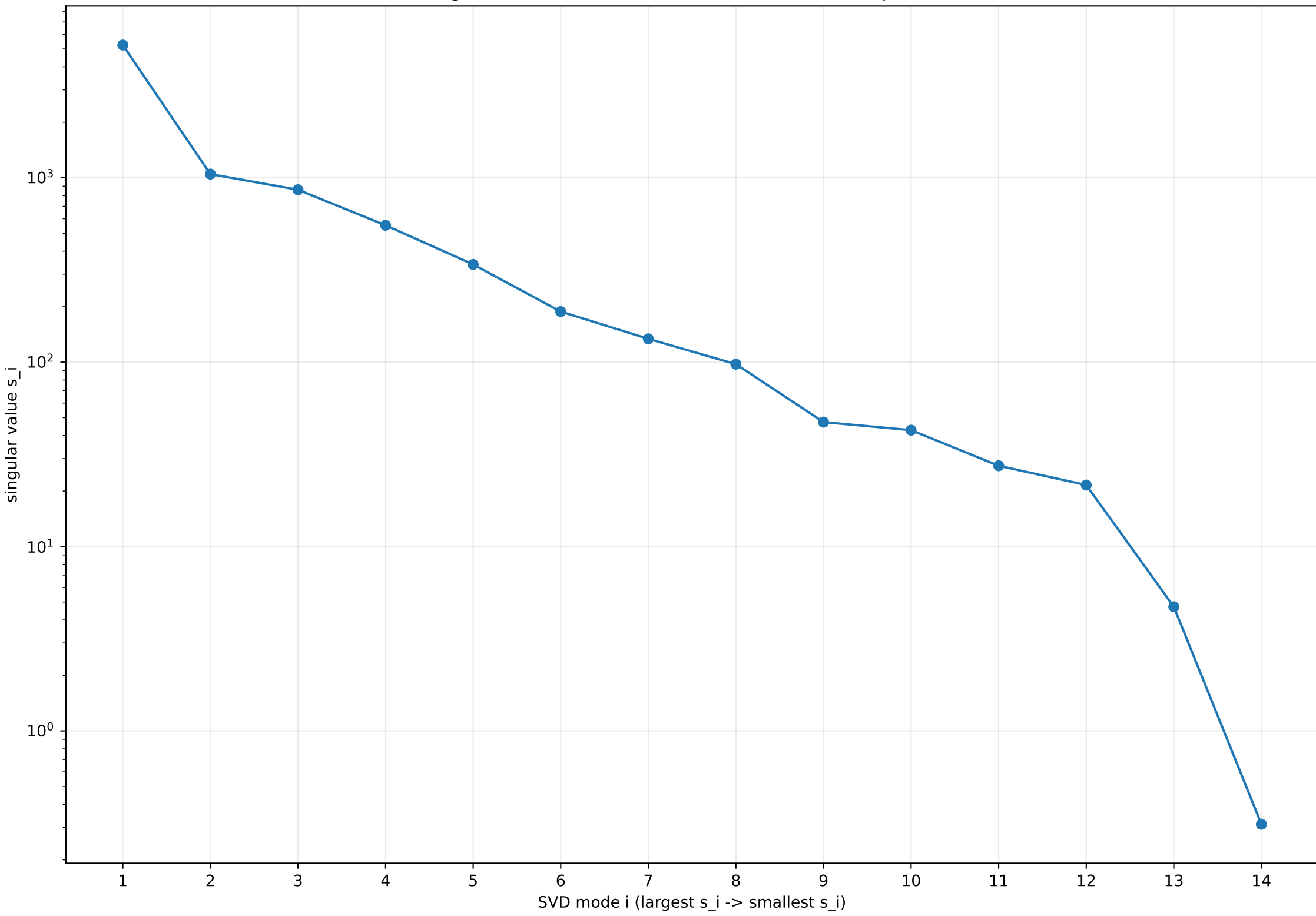


Dimensionless residual torque: empirical mean  $\pm 1$  std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

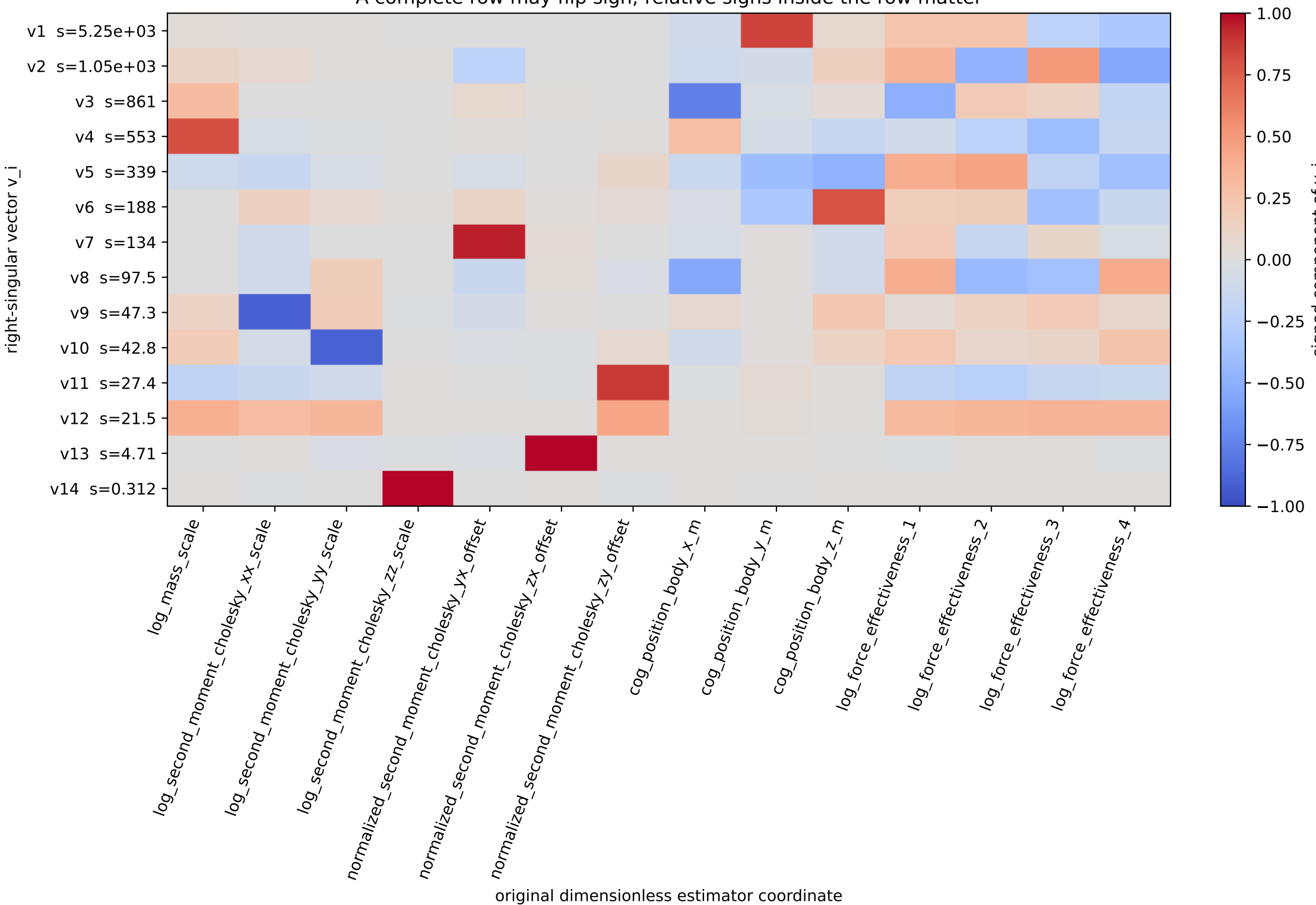




Singular values  $S$  of the whitened dimensionless Jacobian



Signed right-singular vectors  $V^T$   
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36628279309

inertia J [kg m<sup>2</sup>]

[ 0.065011712 -1.0017728e-05 2.3180716e-05]  
[-1.0017728e-05 0.065242046 6.925928e-06]  
[ 2.3180716e-05 6.925928e-06 0.12899475]

principal moments [kg m<sup>2</sup>]

[ 0.065011269, 0.06524248, 0.12899476]

CoG position in body frame [m]

[-0.0026587399, 0.00086274711, 0.010466979]

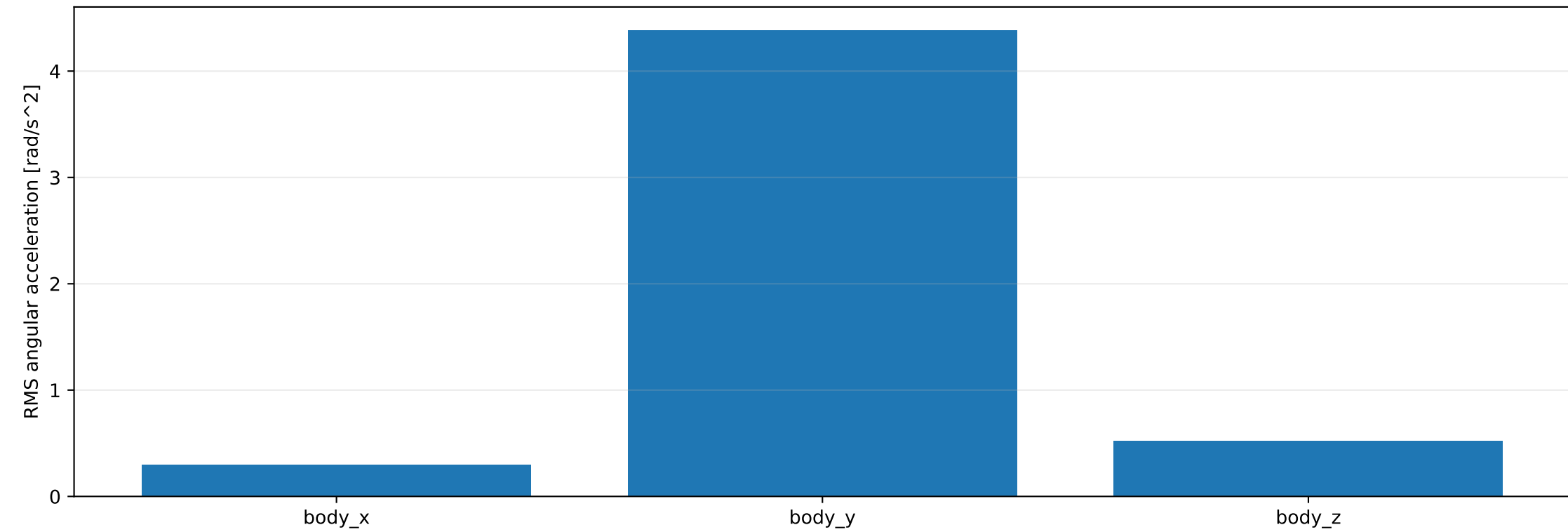
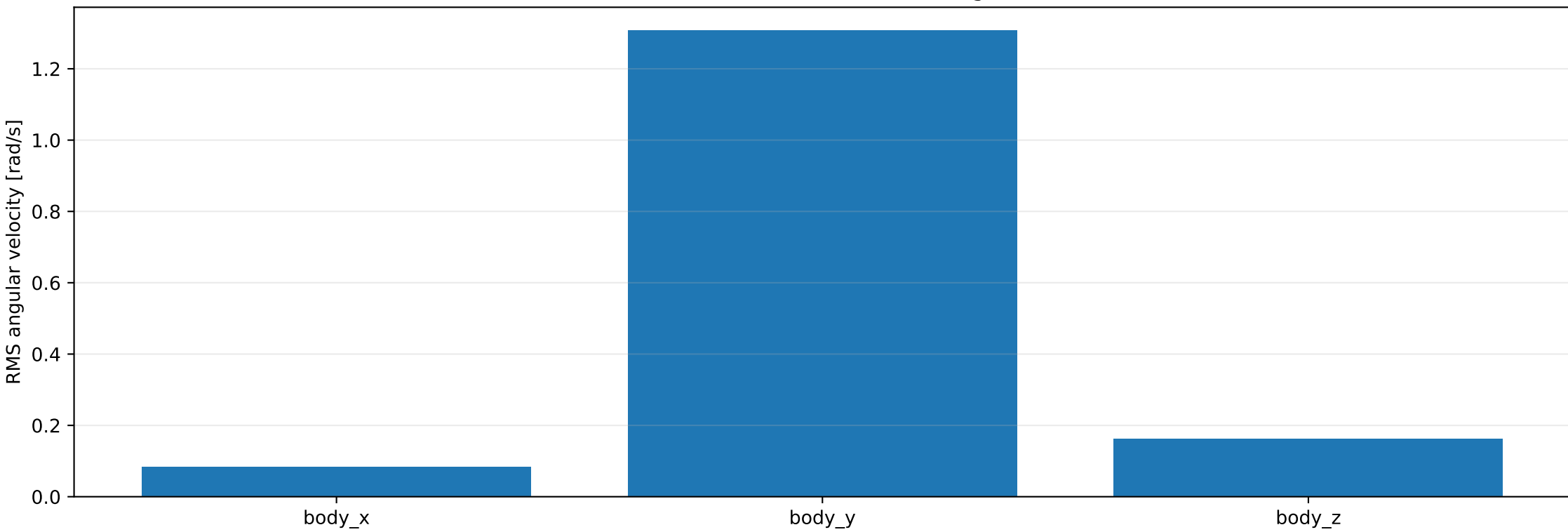
rotor force effectiveness

[ 0.96240339, 0.97340358, 0.95928777, 0.95181648]

selected delay [s]

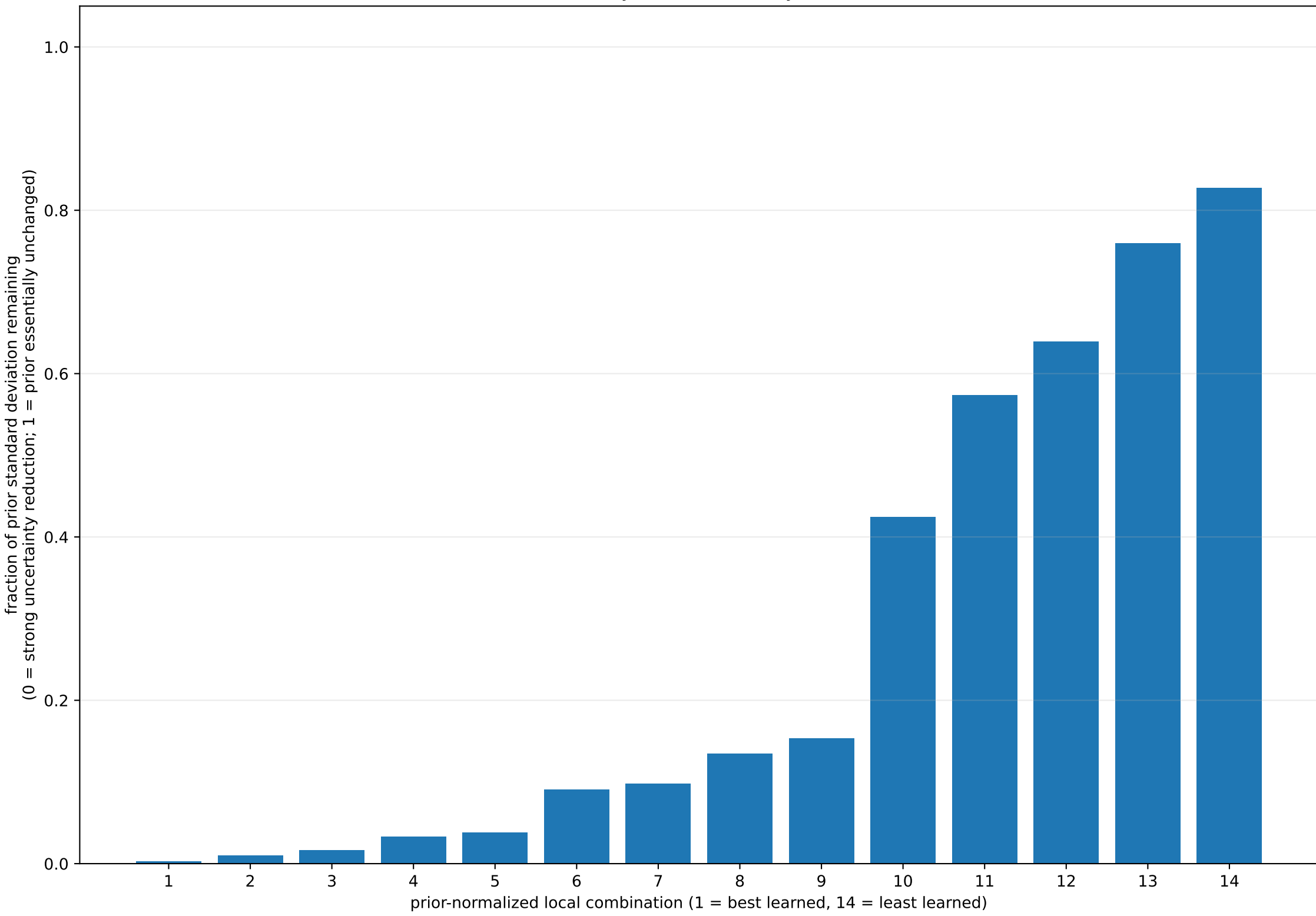
0.195393363886

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

